## The distributive law

**1** The student should multiply the second part of the sum, 2, by the number outside the brackets, 6 to get 12 as the second term on the right hand side.

**2**

a $5s+35$

b $45+9w$

c $3r-18$

d $7y+56$

e $-8t-80$

f $-8c+40$

g $-y-2$

h $16s+20$

i $-80-30r$

j $16t+24$

k $pq+4p$

l $9u-9v$

m $-2m+3n$

n $-11rs-13rt$

o $8r+12s+20t$

p $24m+18n+30p$

**3**

a $12mn+18m$

b $9v^2+54$

c $r^2+7r$

d $24m^2+42m$

e $-12a^2-54a$

f $4wy^2+4w^2y$

g $16pq-12p+24pr$

h $2f^2-14fg-12fh$

i $-10y^2-70$

j $-9+s^3$

k $rs^2+8r$

l $-30-80u^3$

m $16s^5t^4-36s^5$

n $12p^3q^5-54p^3$

o $9f^2-36fg-54fh$

p $3uv^2+3u^2v$

**4**

a $4x+30$

b $15x+27$

c $3x+17$

d $18x+32$

e $36x-11$

f $-11x+24$

g $-x-15$

h $-40x+47$

i $10y-49$

j $20x-52$

**5**

a $x^2+12x+72$

b $15y^2-39y$

c $10x-33y$

d $12y+93$

e $13x-16$

f $4y+44$

g $-11x-26$

h $17a+20$

i $14u^2+20uv$

j $62w^2+34wz$

## Applications

**6** $(11m + 88)$ kW

**7** $(16r - 24)$ representatives

**8** $10 + 15n^4$ boxes

**9** $(16s^2t + 16st^2)$ units$^2$

**10** $(u^2 + 8u)$ m$^2$